

2027 Guide to AI in Science

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2026-08-22

Table of contents

Preface	6
Project Management	8
Project Management: Using AI	9
Top Resources	9
1. <i>[Resource title]</i>	9
2. <i>[Resource title]</i>	9
3. <i>[Resource title]</i>	10
Key Takeaways	10
Points of Disagreement	10
Open Questions	10
Project Management: Critiquing AI	11
Top Resources	11
1. <i>[Resource title]</i>	11
2. <i>[Resource title]</i>	11
3. <i>[Resource title]</i>	12
Key Takeaways	12
Failure Modes to Watch For	12
Points of Disagreement	12
Open Questions	12
Literature Review	13
Literature Review: Using AI	14
Top Resources	14
1. <i>[Resource title]</i>	14
2. <i>[Resource title]</i>	14
3. <i>[Resource title]</i>	14
Key Takeaways	15
Points of Disagreement	15
Open Questions	15

Literature Review: Critiquing AI	16
Top Resources	16
1. <i>[Resource title]</i>	16
2. <i>[Resource title]</i>	16
3. <i>[Resource title]</i>	17
Key Takeaways	17
Failure Modes to Watch For	17
Points of Disagreement	17
Open Questions	17
 Idea Development	 18
Idea Development: Using AI	19
Top Resources	19
1. <i>[Resource title]</i>	19
2. <i>[Resource title]</i>	19
3. <i>[Resource title]</i>	19
Key Takeaways	20
Points of Disagreement	20
Open Questions	20
 Idea Development: Critiquing AI	 21
Top Resources	21
1. <i>[Resource title]</i>	21
2. <i>[Resource title]</i>	21
3. <i>[Resource title]</i>	22
Key Takeaways	22
Failure Modes to Watch For	22
Points of Disagreement	22
Open Questions	22
 Data Collection	 23
Data Collection: Using AI	24
Top Resources	24
1. <i>[Resource title]</i>	24
2. <i>[Resource title]</i>	24
3. <i>[Resource title]</i>	25
Key Takeaways	25
Points of Disagreement	25
Open Questions	25

Data Collection: Critiquing AI	26
Top Resources	26
1. <i>[Resource title]</i>	26
2. <i>[Resource title]</i>	26
3. <i>[Resource title]</i>	27
Key Takeaways	27
Failure Modes to Watch For	27
Points of Disagreement	27
Open Questions	27
 Analysis	 28
Analysis: Using AI	29
Top Resources	29
1. <i>[Resource title]</i>	29
2. <i>[Resource title]</i>	29
3. <i>[Resource title]</i>	29
Key Takeaways	30
Points of Disagreement	30
Open Questions	30
 Analysis: Critiquing AI	 31
Top Resources	31
1. <i>[Resource title]</i>	31
2. <i>[Resource title]</i>	31
3. <i>[Resource title]</i>	32
Key Takeaways	32
Failure Modes to Watch For	32
Points of Disagreement	32
Open Questions	32
 Writing	 33
Writing: Using AI	34
Top Resources	34
1. <i>[Resource title]</i>	34
2. <i>[Resource title]</i>	34
3. <i>[Resource title]</i>	34
Key Takeaways	35
Points of Disagreement	35
Open Questions	35

Writing: Critiquing AI	36
Top Resources	36
1. <i>[Resource title]</i>	36
2. <i>[Resource title]</i>	36
3. <i>[Resource title]</i>	37
Key Takeaways	37
Failure Modes to Watch For	37
Points of Disagreement	37
Open Questions	37
 Dissemination	 38
Dissemination: Using AI	39
Top Resources	39
1. <i>[Resource title]</i>	39
2. <i>[Resource title]</i>	39
3. <i>[Resource title]</i>	39
Key Takeaways	40
Points of Disagreement	40
Open Questions	40
 Dissemination: Critiquing AI	 41
Top Resources	41
1. <i>[Resource title]</i>	41
2. <i>[Resource title]</i>	41
3. <i>[Resource title]</i>	42
Key Takeaways	42
Failure Modes to Watch For	42
Points of Disagreement	42
Open Questions	42
 Personal AI Philosophy Statements	 43
The statements	43
Nicholas A. Coles	43
<i>[Author]</i>	43
Where we agree	43
Where we disagree	44
 References	 45

Preface

Science has always been a social and collective endeavor¹. Furthermore, from the Manhattan Project to the modern AI race, many of the most consequential advances in human knowledge have come not from lone investigators – but from big teams of scientists². In Fall of 2026, faculty, graduate students, and postdocs across multiple departments at the University of Florida leveraged these principles in a Fall 2026 seminar, titled Collective [Artificial] Intelligence in Science. Our goal? Leverage our own collective intelligence to rapidly make sense of one of the most pressing issues in contemporary research:

How is AI changing the nature of science, for better and for worse?

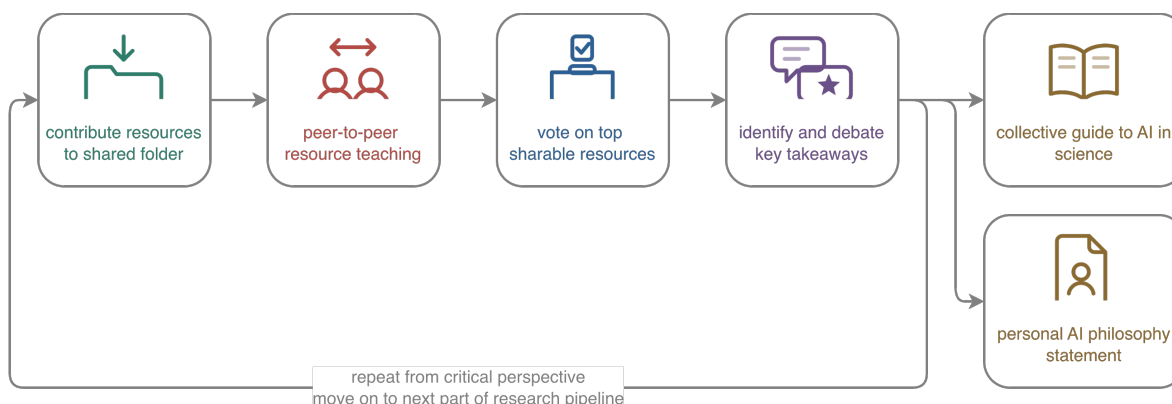


Figure 1: Structure of the University of Florida Fall 2026 seminar, titled Collective [Artificial] Intelligence in Science.

The seminar began with an introduction to big-team science – its evolution, competitive advantages, and relevance to the current moment^{3,4}. From there, the course became a collective investigation, wherein we worked together to understand the newly disrupted nature of each stage of our research pipelines. This includes project management, literature review, idea development, data collection, analysis, writing, and dissemination⁵. To maintain balanced perspective, we alternated between deploying receptive vs. critical lens – identifying key resources, consensus statements, and areas of disagreement along the way. Throughout the semester, we organized our key insights into the resource you are viewing now: the 2027 Guide to AI in Science. This guide was used to teach a MasterClass on AI in Science at the University of Florida — at which time it was also made openly-available for all.

To develop this guide, each week followed a consistent structure (Figure 1). First, we would select a component of the research pipeline to examine (e.g., data analysis). Next, we would spend one week compiling resources for understanding the emerging role of AI in that part of the research pipeline. In class, we engaged in peer-to-peer teaching – all with the goal of identifying the top 3 resources to include in the guide. Last, we used a live online voting platform to solicit, debate, and vote upon key takeaways for the week (<https://pol.is/>). The result is the 2027 Guide to AI in Science: a summary of our top resources, takeaways, and disagreements.

Project Management

Project Management: Using AI

How is AI changing the way research teams plan, coordinate, and keep track of their work?

Lens: receptive — what AI makes newly possible at this stage of the research pipeline.

Top Resources

The three resources the seminar voted to carry forward.

1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

Why it made the cut:

Who it is for:

2. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

3. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Project Management: Critiquing AI

What breaks when the coordination of a research team is delegated to AI?

Lens: critical — the failure modes, costs, and limits of AI at this stage of the research pipeline.

Top Resources

The three resources the seminar voted to carry forward.

1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

The critique it makes:

How strong is the evidence:

2. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

3. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

Failure Modes to Watch For

- *[concrete way this goes wrong in practice, and the signal that it has]*

Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Literature Review

Literature Review: Using AI

How is AI changing the way we find, triage, and synthesize prior work?

Lens: receptive — what AI makes newly possible at this stage of the research pipeline.

Top Resources

The three resources the seminar voted to carry forward.

1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

Why it made the cut:

Who it is for:

2. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

3. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Literature Review: Critiquing AI

What does AI-mediated search hide, distort, or invent — and how would we notice?

Lens: critical — the failure modes, costs, and limits of AI at this stage of the research pipeline.

Top Resources

The three resources the seminar voted to carry forward.

1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

The critique it makes:

How strong is the evidence:

2. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

3. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

Failure Modes to Watch For

- *[concrete way this goes wrong in practice, and the signal that it has]*

Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Idea Development

Idea Development: Using AI

How is AI changing the way research questions and hypotheses are generated?

Lens: receptive — what AI makes newly possible at this stage of the research pipeline.

Top Resources

The three resources the seminar voted to carry forward.

1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

Why it made the cut:

Who it is for:

2. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

3. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Idea Development: Critiquing AI

What happens to originality and intellectual ownership when ideas are co-generated?

Lens: critical — the failure modes, costs, and limits of AI at this stage of the research pipeline.

Top Resources

The three resources the seminar voted to carry forward.

1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

The critique it makes:

How strong is the evidence:

2. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

3. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

Failure Modes to Watch For

- *[concrete way this goes wrong in practice, and the signal that it has]*

Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Data Collection

Data Collection: Using AI

How is AI changing the way we recruit participants, build instruments, and gather data?

Lens: receptive — what AI makes newly possible at this stage of the research pipeline.

Top Resources

The three resources the seminar voted to carry forward.

1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

Why it made the cut:

Who it is for:

2. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

3. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Data Collection: Critiquing AI

What threats to data integrity, validity, and research ethics does AI introduce?

Lens: critical — the failure modes, costs, and limits of AI at this stage of the research pipeline.

Top Resources

The three resources the seminar voted to carry forward.

1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

The critique it makes:

How strong is the evidence:

2. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

3. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

Failure Modes to Watch For

- *[concrete way this goes wrong in practice, and the signal that it has]*

Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Analysis

Analysis: Using AI

How is AI changing the way we write code, model data, and interpret results?

Lens: receptive — what AI makes newly possible at this stage of the research pipeline.

Top Resources

The three resources the seminar voted to carry forward.

1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

Why it made the cut:

Who it is for:

2. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

3. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Analysis: Critiquing AI

Which analytic errors survive when the analysis is delegated?

Lens: critical — the failure modes, costs, and limits of AI at this stage of the research pipeline.

Top Resources

The three resources the seminar voted to carry forward.

1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

The critique it makes:

How strong is the evidence:

2. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

3. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

Failure Modes to Watch For

- *[concrete way this goes wrong in practice, and the signal that it has]*

Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Writing

Writing: Using AI

How is AI changing the drafting, revising, and labor of scientific writing?

Lens: receptive — what AI makes newly possible at this stage of the research pipeline.

Top Resources

The three resources the seminar voted to carry forward.

1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

Why it made the cut:

Who it is for:

2. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

3. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Writing: Critiquing AI

What is lost when writing stops being thinking — and who counts as an author?

Lens: critical — the failure modes, costs, and limits of AI at this stage of the research pipeline.

Top Resources

The three resources the seminar voted to carry forward.

1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

The critique it makes:

How strong is the evidence:

2. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

3. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

Failure Modes to Watch For

- *[concrete way this goes wrong in practice, and the signal that it has]*

Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Dissemination

Dissemination: Using AI

How is AI changing peer review, publishing, and how findings reach their audience?

Lens: receptive — what AI makes newly possible at this stage of the research pipeline.

Top Resources

The three resources the seminar voted to carry forward.

1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

Why it made the cut:

Who it is for:

2. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

3. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Dissemination: Critiquing AI

What does AI do to trust, gatekeeping, and the integrity of the scientific record?

Lens: critical — the failure modes, costs, and limits of AI at this stage of the research pipeline.

Top Resources

The three resources the seminar voted to carry forward.

1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

The critique it makes:

How strong is the evidence:

2. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

3. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

Failure Modes to Watch For

- *[concrete way this goes wrong in practice, and the signal that it has]*

Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Personal AI Philosophy Statements

Having examined the research pipeline together, where does each of us now stand?

The preceding chapters record where the seminar converged as a group. This chapter records where its members stand individually.

Alongside the shared guide, participants wrote personal AI philosophy statements: short, signed accounts of how each of us intends to use — and refuse to use — AI in our own research. Each statement is reproduced in full below and is free to share, quote, or adapt.

The statements

Nicholas A. Coles

[Role, department, institution]

[Statement.]

[Author]

[Role, department, institution]

[Statement.]

Where we agree

Commitments that recur across the statements.

1. *[point of agreement]*
2. *[point of agreement]*
3. *[point of agreement]*

Where we disagree

The substantive splits — not differences of emphasis, but positions that cannot both be right.

- *[position vs. counter-position — and what would settle it]*

Each statement speaks only for its author. Disagreement between them is the point, not an error to be corrected.

References

1. Merton, R. K. *On Social Structure and Science*. (University of Chicago Press, 1996).
2. Coles, N. A., Hamlin, J. K., Sullivan, L. L., Parker, T. H. & Altschul, D. [Build up big-team science](#). *Nature* **601**, 505–507 (2022).
3. Wuchty, S., Jones, B. F. & Uzzi, B. [The increasing dominance of teams in production of knowledge](#). *Science* **316**, 1036–1039 (2007).
4. Coles, N. A., Goes Braga Takayanagi, J. F., Grant, G. L. & Basnight-Brown, D. M. [The rise, impact, and imbalances of big-team psychology](#). *Psychological Science* **37**, 287–297 (2026).
5. .